Collective Radiance:

A Manifestational Ontology of Animal and Insect Group Behavior

Lumen Feng

(冯鲁门)

2026-06

Contents

Copyright Page

Title:

Collective Radiance: A Manifestational Ontology of Animal and Insect Group Behavior

Author:

Lumen Feng（冯鲁门）

ORCID:

https://orcid.org/0009-0003-3716-6557

GitHub:

https://github.com/lumen-feng

DOI:

https://doi.org/10.5281/zenodo.20264872

Version:

Revised Edition 1.0

Publication Form:

Self-published electronic monograph; open access for reading and non-commercial sharing under CC BY-NC-ND 4.0

Completion Date:

June 1, 2026

Theoretical Sources:

This book is developed on the basis of the author's theoretical framework in The Luminous Will: A Manifestational Ontology of Structure, Stability, and Value and From Causal Production to Structural Constraint.

AI Assistance Note：

This book was completed under the direction of a human author and made use of AI tools during writing, revision, structural organization, language refinement, logical checking, and formatting review. AI tools do not constitute the author of this book and bear no intellectual, academic, legal, or ethical responsibility for it. All views, conceptual choices, argumentative arrangements, textual finalization, and any remaining errors are the responsibility of the author alone.

Responsibility Statement：

This book is a theoretical work in the fields of philosophy, philosophy of science, and life systems theory. The analysis of animal and insect group behavior within the book falls under structural and philosophical interpretations and does not replace empirical research in animal behavior, ecology, evolutionary biology, or neuroscience.

Recommended Citation：

Lumen Feng (冯鲁门), Collective Radiance: A Manifestational Ontology of Animal and Insect Group Behavior, Revised Edition 1.0, 2026.

Contact：

lumen_feng@pm.me

Abstract

This book applies the manifestational ontology of the Luminous Will to the analysis of group behavior among animals and insects. Traditional explanations often oscillate between two directions: on the one hand, they reduce group behavior to individual actions, local stimuli, and causal mechanisms; on the other hand, they easily over-totalize group order through expressions such as "collective intelligence," "collective will," or "superorganism." This book proposes a third path: the group behavior of animals and insects is neither a simple sum of isolated individual behaviors nor the appearance of a mysterious collective subject, but a higher-order manifestational structure formed by multiple life-vessels within a shared environment, local signals, feedback mechanisms, and a space of admissibility.

On the basis of the explanatory division among causal production, structural constraint, and admissibility, this book further employs a six-level model: the unmanifest precondition, the persistence of difference, radiance, topology and admissibility, the vessel, and collapse and reorganization. Through this model it analyzes ant trails, bee colonies, termite mounds, locust swarms, bird flocks, fish schools, and herds or packs. The book emphasizes that an individual action is only an occurrence; group behavior truly manifests only when local differences acquire persistence, stable recognizability, and a bearing form within constraint.

The aim of this book is not to propose new biological mechanisms, nor to replace empirical science, but to provide a restrained structural language for centerless order in complex living systems. It acknowledges the importance of individual mechanisms while also showing that group stability cannot be fully compressed into the causal sum of individual actions. It acknowledges the explanatory power of whole patterns while refusing to mystify the group as a supersubject. In this way, group behavior is understood as a higher-order collective vessel formed among life-vessels: it takes shape within constraint, becomes exhausted in time, collapses under disturbance, and reorganizes under new conditions.

Keywords: manifestational ontology; Luminous Will; group behavior; animal behavior; insect societies; structural constraint; admissibility; higher-order vessel; collective intelligence; self-organization; emergence

Preface

Why Return from Civilization to Animal Groups

The Luminous Will: A Manifestational Ontology of Structure, Stability, and Value places life, consciousness, civilization, myth, AI, and value within a broader question: being is not mere occurrence, but continuous manifestation under structural constraint. If this judgment remains only at the level of civilization and value, it can easily be understood as a grand humanistic philosophy. Yet if it is brought back to the group behavior of animals and insects, it obtains a more concrete, more natural, and more comparative field of problems.

Ant colonies form paths; bee colonies maintain divisions of labor; termites build nests; bird flocks turn in synchrony; fish schools evade predators as a whole; herds migrate and protect their young. Many individuals do not possess an overall blueprint, and no central subject issues orders, yet stable, recognizable, maintainable, breakable, and reorganizable group orders still arise. Such phenomena force us to understand anew the source of order: order is not always produced by a center, nor is it always designed by a subject. Many forms of order gradually manifest among local differences, environmental boundaries, signal feedback, and admissible paths.

Therefore, for this book to return from civilization to animal groups is not to lower the theoretical scale, but to seek the natural basis of manifestational ontology. If civilization is a higher-order vessel of radiance on the historical scale, then the group behavior of animals and insects is a prehistoric form of the higher-order vessel within natural living systems. These groups have no linguistically articulated history, no complete institutions, and no reflection on value; nevertheless, they already display structure, boundary, feedback, stability, exhaustion, and reorganization.

The writing of this book will remain restrained. We will not say that an ant colony possesses a mysterious soul, that a bee colony is a supernatural subject, or that the synchrony of a bird flock is an ineffable cosmic will. Instead, this book attempts to show that acknowledging the whole structure of a group does not mean mystifying the whole, and that opposing mystical holism does not require compressing everything into the causal sum of isolated individuals. The meaning of manifestational ontology lies precisely in the intermediate path it provides.

A group is not commanded into being. A group takes shape within constraint.

Introduction

How Order Manifests without a Central Subject

The most fascinating aspect of animal and insect group behavior is that it often presents an order that appears whole. When a flock turns, it resembles an immense flowing body; when a school of fish evades a predator, it resembles an elastic boundary; when ants forage, they form a continually renewed path network; when bees divide labor, they form a complex system capable of regulating temperature, storing resources, reproducing, and migrating. Yet when we ask, "Who controls all this?" we often cannot find a single answer.

Traditional explanations can of course identify many concrete mechanisms: pheromones, visual distance, neighbor responses, division of roles, hormonal states, environmental pressure, and evolutionary selection. Without these mechanisms, group behavior could not be concretely explained. But the question does not stop there. Knowing how a certain behavior occurs is not the same as explaining how a whole form becomes stable; knowing how a single individual responds to a stimulus is not the same as explaining how a collective vessel takes shape.

The basic question of this book can be stated as follows: in the absence of a central subject, how does group order manifest from local behavior? This is not an anti-scientific question but a question of explanatory level. Causal production tells us how a certain action occurs; structural constraint tells us which paths are opened or excluded; admissibility tells us which group forms can be borne by the system; manifestational theory further asks what whole form acquires stable recognizability when these conditions operate together.

Therefore, this book will not treat the phrase "collective intelligence" as a final answer; rather, it will decompose it. So-called collective intelligence does not mean that a hidden mind stands behind the group. It means only that multiple living individuals, within local rules, a shared environment, and feedback mechanisms, form a higher-order operable order. When this order can be recognized, maintained, functionally used, and reorganized after disturbance, it is no longer mere occurrence; it has manifested.

Part I

From Individual Behavior to Collective Manifestation

The task of Part I is to develop an applied analysis of manifestational theory around the movement from individual behavior to collective manifestation. The following chapters explain, through concepts, levels, cases, and boundaries of value, how group behavior obtains stable recognizability within structural constraint.

Chapter 1

Why Group Behavior Is a Touchstone for Manifestational Theory

Core Proposition of This Chapter

Group behavior matters because it shows most clearly that stable and recognizable whole order can emerge even without a central subject.

1.1 The Basic Difficulty of Group Behavior

The group behavior of animals and insects has philosophical significance because it places a deep problem before us: single individuals do not possess an overall blueprint, yet they are able to form whole order. An ant colony does not draw a complete map; termites do not first draw architectural plans; no bird stands at the center of a flock to issue commands; no fish school has a shared consciousness that calculates collective motion. Nevertheless, they can still form paths, nests, formations, boundaries, and migration structures.

This phenomenon can be reduced neither simply to the sum of individual behaviors nor hastily to a mysterious whole will. The former cannot see the stability of the whole form; the latter hands explanation over to an unrestricted anthropomorphic imagination. This book treats group behavior as a touchstone for manifestational theory precisely because it requires us to acknowledge both local mechanisms and whole forms.

1.2 Why We Cannot Speak Only of Instinct

The word "instinct" can describe a hereditary basis or tendency of behavior. But if all group order is compressed into instinct, the structural problem is concealed. To say that ants instinctively forage does not explain why one path becomes stable; to say that birds instinctively flock does not explain why a group maintains distance, direction, and boundary while moving.

Instinct may be a condition of individual behavior, but it does not automatically explain how a collective vessel takes shape. Manifestational theory is concerned with the following questions: which differences persist? Which feedback loops are amplified? Which paths are excluded? Which whole patterns are borne by the system? These questions require an analysis of structural constraint and admissibility.

1.3 Why We Cannot Mystify It as a Collective Soul

Another common tendency is to imagine complex group order as if a unified mind stood behind the group. When people see a bird flock turning like a whole, they say it resembles an enormous living body; when they see the division of labor and temperature regulation in a bee colony, they say it resembles a superorganism. These metaphors can be illuminating, but if they lose their limits, they slide into mystification.

Manifestational theory acknowledges that the whole of the group has explanatory significance, but it does not reify the whole into a soul. A group is not a hidden person; it is a higher-order bearing structure formed among multiple life-vessels in relation. It can stand above the individual without having to become a new subject.

1.4 The Third Path of Manifestational Theory

Manifestational theory proposes a third path between individual reductionism and mystical holism: group behavior is the process in which local differences acquire persistence, stable recognizability, and a bearing form within structural constraint. This process requires concrete mechanisms, yet it cannot be completely replaced by those mechanisms.

The behavior of a single individual is only an occurrence. When such occurrences are organized by signals, distance, boundaries, environment, and feedback, and when they form a recognizable whole pattern, group behavior truly manifests. Collective radiance is not mystical light, but the stable recognizability of a collective form.

1.5 The Object and Method of This Book

This book will analyze ant trails, bee colonies, termite mounds, locust swarms, bird flocks, fish schools, and herds or packs. They respectively display different types of collective vessels: path, division of labor, architecture, negative grouping, dynamic synchrony, fluid boundary, and social memory.

This book does not replace ethology, does not propose new empirical mechanisms, and does not deny the importance of evolutionary, ecological, neural, and behavioral mechanisms. Its contribution is to provide a philosophical language for how these mechanisms are organized into higher-order collective forms.

Chapter Summary

This chapter establishes animal and insect group behavior as a concrete touchstone for manifestational theory. Ant colonies, bee colonies, bird flocks, and fish schools show that stable whole order can be generated by local relations without a central subject. The key is not to posit a hidden collective soul, but to analyze how differences are continuously maintained by signals, environment, distance, and feedback.

Group behavior therefore provides a natural entry point into manifestational theory. It preserves the importance of individual mechanisms while forcing us to face the stability, boundary, bearing capacity, and recognizability of the whole form.

Chapter 2

From Causal Production to Structural Constraint: The Division of Explanation in Group Behavior

Core Proposition of This Chapter
Causal production can explain how a single behavior occurs, but the stability of group order also requires explanations through structural constraint and admissibility.

2.1 What Causal Production Explains

Causal production explains how a certain behavior occurs. For example: why an ant releases pheromones, why a bee performs a certain role, why a bird adjusts in a particular direction, or why a fish accelerates or turns when a predator approaches. These questions all require concrete mechanisms. Without explanations of causal production, group behavior would become a vague whole-effect.

However, knowing how a certain action occurs is not the same as explaining how whole order becomes stable. A single ant releasing pheromone is a causal process; an entire ant trail stabilizing into form is a structural process. A single bird turning is a local response; the whole flock turning involves neighbor relations, speed, distance, and temporal delay.

2.2 What Structural Constraint Explains

Structural constraint answers which paths, states, and forms are limited or excluded. Group behavior always occurs within constraint. Ant trails are constrained by terrain, pheromone evaporation, food returns, and nest location; bee colonies are constrained by nest space, role structure, and resource flow; bird flocks are constrained by collision-avoidance distance, speed synchronization, and field of vision.

Constraint is not the enemy of group order but its condition. Without distance constraints, a bird flock would either scatter or collide; without signal constraints, an ant trail could not stabilize; without nest boundaries, the division of labor in a bee colony would be difficult to bear.

2.3 What Admissibility Explains

Admissibility answers the following question: under given structural conditions, which group patterns can be stably borne by the system? Not every living system can form every kind of group structure. Ants can form path networks but cannot form linguistically articulated institutions; birds can form dynamic formations but cannot preserve texts; bees can form complex divisions of labor but cannot establish reflective history.

Admissibility is therefore not abstract possibility, but possibility opened by structure. Manifestational theory requires us to ask: what kinds of group forms can be stably sustained by the body, perception, signals, environment, and feedback of this system?

2.4 Group Stability Cannot Be Replaced by Repeated Production

If we ask only what produces these behaviors again and again, we easily misdescribe group stability as repeated production. Yet stability is not the repetition of the same behavior many times. It is the ability of a form to maintain itself amid variation, deviation, and disturbance.

An ant trail may contain individual deviations and still maintain a path; a bird flock may contain positional fluctuations and still maintain its formation; a bee colony may lose individual bees and still preserve division of labor. Stability comes from the structure's capacity to accommodate difference, not merely from the copying of actions.

2.5 The Necessity of Multiple Explanatory Roles

A mature explanation of group behavior requires at least three roles: causal production explains how actions occur; structural constraint explains which paths are opened or excluded; admissibility explains which whole forms can be stably borne. Manifestational theory then explains how the whole form obtains recognizability on this basis.

This division of explanatory roles allows us to avoid two thin explanations: one that looks only at individual mechanisms, and another that speaks only of collective intelligence. True group order lies between the two and is constituted jointly by local mechanisms and higher-order structure.

Chapter Summary

This chapter distinguishes three explanatory roles in group behavior. Causal production explains how individual actions occur; structural constraint explains which paths are opened or excluded; admissibility explains which whole patterns can be stably borne by the system.

Thus, group order is no longer merely individual behavior repeatedly produced; it is stable manifestation formed through the interweaving of multiple explanatory roles. Analysis of group behavior must preserve both mechanism-level detail and higher-order structure.

Chapter 3

The Life-Vessel and the Collective Vessel

Core Proposition of This Chapter
A collective vessel is a higher-order manifestational structure formed by multiple life-vessels within a shared environment, local signals, feedback mechanisms, and boundary conditions.

3.1 Why Life Is Already a Vessel

Life is not an arbitrary pile of matter, but a manifestational structure capable of maintaining its own boundary, exchanging energy and information, repairing damage, and resisting collapse. Life is a vessel because it not only occurs, but continuously bears itself.

Cells, bodies, nervous systems, and behavioral patterns all participate in the maintenance of the life-vessel in different ways. The manifestation of life is not completed once and for all, but maintains local stability through continuous dissipation and repair.

3.2 From the Life-Vessel to the Collective Vessel

When multiple life-vessels enter a shared environment and form a relatively stable whole structure through signals, distance, roles, feedback, and boundaries, the collective vessel begins to appear. The collective vessel does not cancel the individual; rather, it organizes relations among individuals into a bearing form.

Ant trails, beehives, bird flock formations, fish school boundaries, and migration routes are all different types of collective vessels. They are not the same thing, but they share one structural feature: they allow the local actions of multiple individuals to form a recognizable whole.

3.3 The Group Is Not a Large Individual

The collective vessel cannot be simply understood as a large individual. Individuals have bodily boundaries and internal metabolism; groups maintain themselves through relational boundaries, signal exchange, and environmental feedback. A group may display life-like stability, but this stability is distributed.

Therefore, calling a bee colony a superorganism can be a metaphor, but it cannot cancel analysis. Manifestational theory prefers to say that the bee colony has higher-order self-maintaining functions at certain levels, but these functions arise from structural relations among multiple life-vessels.

3.4 The Group Is Not an Ordinary Set

An ordinary set merely places many individuals together. A collective vessel requires a sustainable relational structure among individuals. Many birds standing on the same patch of ground do not necessarily form a flock; many ants moving randomly do not necessarily form an ant trail.

The minimal condition of a collective vessel is that local behaviors can form whole function through signals or relations. Without such functional bearing, aggregation is only an increase in number, not manifestation as a group.

3.5 Types of Collective Vessel

A collective vessel may be temporary, like an ant trail; relatively enduring, like a beehive; dynamic, like a bird flock formation; fluid, like a fish school boundary; materialized, like a termite mound; or mnemonic, like a migration route.

Together these types show that a vessel is not a fixed object, but a bearing relation. As long as a structure can bear collective radiance, maintain boundaries, and face exhaustion, it can be analyzed as a vessel.

Chapter Summary

This chapter understands individual life as a life-vessel capable of maintaining boundaries, exchanging energy, repairing damage, and resisting collapse. On this basis it proposes the concept of the collective vessel. The collective vessel does not cancel individuals, but lets relations among individuals enter a bearing structure.

The group is therefore neither an ordinary set nor a large-scale subject. It is a higher-order bearing structure formed by multiple life-vessels within a shared environment, signals, roles, and feedback.

Part II

The Six-Level Model of Collective Manifestation

The task of Part II is to develop an applied analysis of manifestational theory around the six-level model of collective manifestation. The following chapters explain, through concepts, levels, cases, and boundaries of value, how group behavior obtains stable recognizability within structural constraint.

Chapter 4

L0-L1: The Unmanifest Precondition of Group Behavior and the Persistence of Difference

Core Proposition of This Chapter

The starting point of group order is not a central command, but the persistence of a local difference within environment and feedback.

4.1 What Exists before Group Behavior Appears

Before group behavior manifests, there is not simply nothing. There are environments, resources, threats, individual density, bodily capacities, perceptual ranges, terrain boundaries, and potential signals. They have not yet formed group order, but they already constitute a background that can manifest.

For example, before food is discovered, the ant trail has not yet appeared; before a predator approaches, the fish school has not yet entered a defensive form; before a change in wind direction, the bird flock has not yet adjusted its formation. These are unmanifest preconditions, not absolute nothingness.

4.2 The Unmanifest Is Not Nonexistence

The unmanifest is not "nonexistent"; it is not yet formed. The absence of a group pattern does not mean that all conditions are absent. On the contrary, many conditions are already present, but they have not yet been organized by a difference.

This distinction is important. It prevents us from treating group behavior as if it suddenly jumped out of nothing; instead, we see it as gradually taking shape within an existing background. The emergence of a collective vessel is the result of the encounter between background conditions and a local difference.

4.3 How Local Difference Is Opened

A difference can be very small. One ant finds food, one bird changes direction, one fish senses danger, one bee performs a certain role: at first, these are all local events. Whether they become group order depends on whether the difference persists.

Difference opens the first fissure of manifestation. Without difference, the group remains an undifferentiated background; if the difference immediately disappears, no collective radiance will form.

4.4 Will in the Group Is Not Psychological Will

When this book uses the word "will" in relation to group behavior, its meaning must be strictly limited. Collective will is not a group mind, not a whole desire, and not a hidden person. It is only the functional name for the persistence of difference.

When a path difference is reinforced by pheromones, when a directional difference is answered by neighboring individuals, when a migration route is repeated across generations, difference has not returned to the unmanifest. This "not returning" is will within collective manifestation.

4.5 The Persistence of Difference as the Starting Point of Group Order

Group order does not begin from a grand plan. It begins from the persistence of a local difference. This starting point is modest, yet sufficient to explain many forms of centerless order.

As long as a difference can be preserved by the environment, answered by other individuals, and reinforced by feedback, it may enter the next level: from the persistence of difference to the stable and recognizable collective radiance.

Chapter Summary

This chapter shows that before group order appears, there is always an unmanifest background: resources, threats, density, terrain, perceptual capacities, and signal conditions jointly constitute a field of possibility. Only when a local difference is opened and persists does collective manifestation truly begin.

Here, "will" is not a mysterious psychological will of animals or insects, but the functional name for a difference that does not immediately disappear. Pheromone paths, directional deviations, alarm signals, and role tendencies are concrete entries into the persistence of collective difference.

Chapter 5

L2-L3: Collective Radiance, Topology, and the Space of Admissibility

Core Proposition of This Chapter
Collective radiance is the stable and recognizable whole pattern formed by multiple living individuals within structural constraint.

5.1 Collective Radiance Is Not Mystical Light

Collective radiance is neither physical light nor mystical energy. It refers to the stable recognizability acquired by a group pattern. An ant trail has radiance because random movement is no longer merely random movement, but has formed a discernible path; a bird flock has radiance because individual motions have formed a whole formation.

Radiance is not an ornament added to the group, but the recognizability through which the group form appears. Without stable recognizability, group behavior is only a heap of events; with stable recognizability, the group begins to become an analyzable form.

5.2 From Random Motion to Recognizable Formation

In many forms of group behavior, randomness does not completely disappear. Ants deviate from paths, birds change positions, and fish undergo local disturbance. Yet the whole form may still be maintained. This shows that collective radiance is not absolute regularity, but the maintenance of contour while accommodating difference.

Recognizability does not require the absence of change. On the contrary, mature collective radiance can often withstand change. It is not a rigid structure, but a relational form capable of maintaining itself amid fluctuation.

5.3 Topology: Distance, Direction, Role, and Relation

Topology is the structure of relations within the group. Distance and direction in a bird flock, boundary and density in a fish school, role distribution in a bee colony, and path networks in an ant colony are all topologies. Topology determines how difference propagates, how feedback accumulates, and how local behavior affects the whole.

If we ignore topology, we can see only individual behavior, but we cannot see why those behaviors can organize into a whole. Topology is not an abstract background; it is an actual condition within the process of manifestation.

5.4 Admissibility: What Kind of Group Can Become Stable

Admissibility indicates which group forms a given system can stably bear. Different species have different bodies, perceptions, modes of movement, and environmental relations, and therefore their spaces of admissibility differ.

The admissibility of an ant colony is shaped by scent paths, surface structures, and food returns; the admissibility of a bird flock is shaped by flight speed, visual field, collision avoidance, and wind direction; the admissibility of a bee colony is shaped by the nest, division of labor, and reproductive structure.

5.5 Collective Radiance Depends on Constraint

Collective radiance does not appear after escaping constraint; it appears within constraint. Without boundaries, a fish school cannot form a defensive contour; without role relations, a bee colony cannot maintain its division of labor; without path feedback, an ant trail cannot stabilize.

Constraint is therefore not the opposite of manifestation, but its condition. Collective radiance becomes visible because certain paths are excluded, certain relations are maintained, and certain forms are borne by the system.

Chapter Summary

This chapter defines collective radiance as the stable recognizability of a group pattern. Ant trails, bird flock formations, fish school boundaries, and bee-colony divisions of labor can be seen because local behavior has formed a discernible contour within topological relations.

Admissibility further shows that not every group form can stably exist. Bodies, perception, signals, distance, environment, and feedback together open certain paths and exclude others, allowing collective radiance to appear within concrete limits.

Chapter 6

L4-L5: The Collective Vessel, Exhaustion, Collapse, and Reorganization

Core Proposition of This Chapter
The collective vessel is not an eternal order, but a higher-order manifestational structure that maintains local stability through exhaustion, rupture, feedback, and reorganization.

6.1 How the Collective Vessel Forms

When collective radiance obtains a bearing form, the collective vessel is formed. An ant trail bears foraging; a beehive bears reproduction and storage; a termite mound bears temperature and humidity regulation; a bird flock formation bears movement synchrony; a fish school boundary bears defense.

The key to the vessel is not whether it is hard, but whether it bears. A dynamic formation can be a vessel; a migration route can be a vessel; a temporary path can be a vessel.

6.2 The Vessel Is Not a Static Thing

Collective vessels are often more fluid than ordinary objects. A bird flock formation exists in movement; a fish school boundary changes in the current; an ant trail shifts with the strength of pheromones; a bee colony adjusts its division of labor with seasons and resources.

Thus, the vessel is not a static shape, but a bearing relation. As long as a relation can maintain a boundary, organize function, and remain relatively stable over time, it has the character of a vessel.

6.3 Why the Collective Vessel Becomes Exhausted

Every vessel has a boundary, and everything with a boundary has a limit of bearing. Collective vessels become exhausted through resource depletion, signal decay, environmental destruction, disease, predation, individual death, or role imbalance.

An ant trail disappears after food is depleted; a bee colony collapses under disease or environmental pressure; a bird flock breaks under strong disturbance; a fish school disperses under predatory pressure. These are not theoretical failures, but the finitude of the vessel.

6.4 Collapse Is Not Pure Failure

Collapse does not always mean that collective radiance disappears completely. A formation can reorganize after breaking; after one ant trail disappears, a new path can appear; after a bee colony swarms, a new nest can form.

Manifestational theory treats collapse as part of the process of manifestation. A vessel becomes exhausted, but differences, memories, residual paths, and environmental conditions may provide the background for new manifestation.

6.5 The Meaning of Local Stability

Group stability is never absolute stability, but local stability. It holds within specific times, spaces, and conditions. The more complex a collective vessel is, the more it must continually revise itself.

This allows us to understand stability anew: stability is not changelessness, but the maintenance of recognizability within change; it is not the elimination of disturbance, but the capacity to absorb disturbance, adjust boundaries, and reorganize structure.

Chapter Summary

This chapter understands the collective vessel as a stable structure capable of bearing collective function. Ant trails, beehives, bird flock formations, fish school boundaries, and termite mounds are not isolated events, but bearing forms that maintain themselves for a certain time.

Yet the collective vessel inevitably faces exhaustion. Food disappears, signals decay, environments are disturbed, and individuals die. The maturity of group order does not consist in eternal immutability, but in the capacity to reorganize local stability after rupture.

Part III

Insect Societies: From Paths to Nests as Collective Vessels

The task of Part III is to develop an applied analysis of manifestational theory around insect societies: from paths to nests as collective vessels. The following chapters explain, through concepts, levels, cases, and boundaries of value, how group behavior obtains stable recognizability within structural constraint.

Chapter 7

Ant Trails: The Persistence of Difference and the Temporary Path-Vessel

Core Proposition of This Chapter
The ant trail is not the result of central planning, but the result of a local path difference that persists into a vessel through pheromones, terrain, food return, and group feedback.

7.1 Why Ant Foraging Is Suited to Manifestational Analysis

Ant foraging is a classical scene for understanding collective manifestation. The exploration of a single ant may be random, but after a food path is discovered and reinforced by pheromones, random exploration gradually converges into a stable route.

Here there is no central planner, yet there is path order; there is no whole map, yet there is an efficient route. This makes the ant trail the first natural example for manifestational analysis.

7.2 The Movement of a Single Ant Is Only an Occurrence

The movement of a single ant is, in itself, only an occurrence. It may go left, go right, return, or deviate. This local behavior is not yet a collective vessel.

Only when this movement leaves a trace and is answered, repeated, and reinforced by other ants does local behavior cross beyond a single occurrence and enter the persistence of difference.

7.3 How Pheromones Make Difference Persist

Pheromones can be understood as a mechanism for preserving difference in the environment. They make a path no longer merely a place where one individual has passed, but a trace readable by later individuals.

Pheromones are neither commands nor maps. They are more like a medium of local manifestation, allowing path differences to acquire persistence in the environment.

7.4 The Ant Trail as a Temporary Collective Vessel

When many ants act along the same path, the path becomes a temporary collective vessel. It bears food transport, directional choice, and group feedback. Its boundary is not hard, yet it is sufficient to organize behavior.

The vessel-character of the ant trail lies in bearing, not in material fixity. It is jointly constituted by scent, terrain, bodily action, and repeated feedback.

7.5 Path Exhaustion and Reorganization

After food is exhausted, the original ant trail declines. Pheromones evaporate, following decreases, and the path-vessel gradually loses its bearing capacity.

But this disappearance is not absolute failure. New exploration continues; new differences may be opened; new paths can manifest again. The ant trail shows both the temporariness and reorganizability of the collective vessel.

Chapter Summary

This chapter uses ant foraging to show the formation of a temporary path-vessel. The exploration of a single ant is only an occurrence; when a food path is marked by pheromones, followed by subsequent individuals, and continuously reinforced, the path difference becomes a stable ant trail.

The value of the ant trail lies precisely in its temporariness and replaceability. After food is exhausted, the old path dissipates and a new path manifests. Collective memory is not stored in a central mind, but is temporarily distributed among environmental traces and behavioral feedback.

Chapter 8

Bee Colonies: Division of Labor, the Nest, and Higher-Order Self-Maintenance

Core Proposition of This Chapter
A bee colony is not the sum of individual bees, but a higher-order self-maintaining vessel formed through division of labor, the nest, reproductive structure, resource flow, and signal systems.

8.1 Why a Bee Colony Is Not an Ordinary Set

A bee colony is not the simple addition of many bees. The hive, the queen, workers, drones, larvae, honey storage, temperature regulation, and signal exchange together constitute a higher-order bearing system.

Within this system, the life activity of single individuals is organized into group maintenance. Individuals may die or be replaced, while the group structure may still be maintained.

8.2 The Hive as Boundary

The hive is an important boundary of the bee colony as vessel. It provides not only physical space, but also organizes temperature, reproduction, food storage, and division of labor. Without the nest boundary, the division of labor of the bee colony would be difficult to sustain.

The hive shows that a collective vessel can possess strong materiality. It condenses the actions of many individuals into a sustainable spatial structure.

8.3 Division of Labor and Role Topology

Roles in a bee colony are not mere labels, but group topology. The queen, workers, drones, and the tasks of workers at different stages together constitute the colony's space of admissibility.

Role topology allows the group to distribute resource gathering, brood care, defense, cleaning, nest-building, and temperature regulation among different functional positions. Division of labor is not dispersion, but higher-order unity.

8.4 Signals and Resource Paths

Bees transmit information about resource direction, nest conditions, and group needs through signals. These signals are not abstract language, yet they are sufficient to turn local discovery into group action.

A resource path is therefore not only a spatial path, but also a group information path. The collective vessel of the bee colony depends on such paths to incorporate external resources into internal maintenance.

8.5 Swarming and Reorganization

Swarming is an important form of replication and reorganization for the bee colony as collective vessel. After a group reaches a certain bearing state, it may split into a new collective vessel.

This shows that collective exhaustion does not lead only to death. Certain forms of exhaustion or saturation can lead to reorganization, allowing radiance to continue appearing through a new vessel.

Chapter Summary

This chapter uses the bee colony to show the collective vessel of higher-order self-maintenance. The hive provides a boundary, division of labor provides structure, and resource flow, brood care, temperature regulation, and defense jointly sustain the continuity of the group. The bee colony is therefore no longer merely a collection of individual bees.

The manifestation of the bee colony also appears in swarming and migration. A collective vessel can copy, divide, or reorganize its bearing structure, making the bee colony an important case in the passage from the life-vessel to the social collective vessel.

Chapter 9

The Termite Mound: The Materialization of the Collective Vessel

Core Proposition of This Chapter

The termite mound is not the work of individual intention, but a materialized collective vessel formed by local behavior within environmental feedback and material constraint.

9.1 The Philosophical Significance of Termite Nest-Building

The termite mound matters because it condenses group behavior into visible architecture. Individual termites do not need to possess a complete blueprint, yet local carrying, material accumulation, humidity, air flow, and chemical signals can jointly form a complex structure.

This makes the termite mound a strong example of the vessel: collective radiance is not only a pattern of movement; it can also sediment into material structure.

9.2 Local Behavior and Environmental Feedback

Termite nest-building does not execute a pre-existing plan. It gradually takes shape through local behavior and environmental feedback. Material accumulation at one location changes local conditions, and those local conditions influence later carrying.

This feedback allows the architectural structure to avoid dependence on central design. It is the result of the environment participating in manifestation.

9.3 Humidity, Air Flow, and Structural Constraint

The termite mound is not arbitrary accumulation. Temperature, humidity, gas exchange, and material stability jointly limit which structures can form. Here, structural constraint is not an abstract concept, but the actual condition for whether the nest can bear group life.

If ventilation cannot hold, the nest cannot maintain itself; if the materials are unstable, the structure will collapse; if the spatial boundary loses balance, group function will be damaged.

9.4 Materialized Collective Radiance

Once a termite mound has formed, collective radiance obtains a materialized form. It is not merely a trace of termite behavior, but a vessel that continues to organize termite life.

The nest is both result and condition. It is generated by group behavior and in turn shapes group behavior.

9.5 Damage, Repair, and Reorganization

A termite mound can be damaged, and it can also be repaired. Repair is not simply a restoration of the original form, but a reorganization of the bearing structure under existing damaged conditions.

The termite mound therefore displays the materialization, exhaustion, and reorganization of the collective vessel. It shows that a vessel does not take shape once and then remain unchanged; it manifests through continuous maintenance.

Chapter Summary

This chapter shows that the termite mound is a materialized form of collective manifestation. Individual termites need not possess an overall architectural blueprint; local carrying, material accumulation, humidity, temperature, and air-flow feedback are already sufficient to drive the gradual formation of structure.

The termite mound shows that the collective vessel can move beyond pure behavioral patterns and enter architectural bearing. The nest is both a product of group activity and something that in turn shapes the group's space of admissibility.

Chapter 10

Locust Swarms: The Negative Form of Collective Manifestation

Core Proposition of This Chapter
Collective manifestation can be destructive; the ability to form a stable collective vessel does not mean that the structure has the good in an ethical sense.

10.1 Why Locust Swarms Must Be Discussed

If we discuss only ant trails, bee colonies, bird flocks, and fish schools, we easily misunderstand collective manifestation as naturally positive order. Locust swarms force us to face another side: a group structure can be effective, coordinated, and powerful while also being destructive.

Manifestation is not a value judgment. The fact that a structure can form does not mean that it deserves continuation; the fact that a group can expand does not mean that it has ethical legitimacy.

10.2 The Negative Collective Vessel

Under particular densities, resources, and environmental pressures, locusts form aggregation, migration, and large-scale feeding behavior. There is persistence of difference, collective radiance, and a moving vessel.

Yet the main effect of this vessel may be the devouring of resources and the transfer of environmental pressure. Since it maintains itself by consuming external bearing conditions, it can be understood as negative collective manifestation.

10.3 Stability Is Not the Good

Many natural structures can exist stably, but they do not thereby become good. Predation, parasitism, plunder, and outbreaks can all appear stably in nature.

If natural selection or adaptive success is directly transformed into an ethical standard, we fall into the naturalistic fallacy. Manifestational theory must separate stability from value.

10.4 The Limit of Destructive Radiance

The radiance of a locust swarm is real, but its direction is dissipative. It can manifest, yet it limits future complex manifestation by consuming bearing conditions.

This reminds us that not all radiance deserves praise, and not all vessels deserve continuation. A theory of value must ask about the direction of the structure, not merely acknowledge its strength.

10.5 Its Relation to Civilizational Value Theory

At the natural level, the locust swarm poses a question that also applies to civilization: if a powerful structure maintains itself primarily by devouring, destroying, and consuming the future, should it be called good?

The answer of this book is no. The good is not strength itself, but the structural direction that enhances non-destructive, sustainable, complex manifestation.

Chapter Summary

This chapter treats the locust swarm as a case of negative collective manifestation. The locust swarm likewise possesses stability, directionality, and collective scale, but this stability is not automatically equivalent to the good; it is often characterized by resource consumption, destructive spread, and the exhaustion of bearing conditions.

Manifestational theory must therefore distinguish natural stability from ethical value. A group structure that can take shape, expand, or persist does not necessarily have the meaning of the good. The question of value must separately examine whether it enhances sustainable complex bearing.

Part IV

Animal Groups: Centerless Synchrony and Moving Vessels

The task of Part IV is to develop an applied analysis of manifestational theory around animal groups: centerless synchrony and moving vessels. The following chapters explain, through concepts, levels, cases, and boundaries of value, how group behavior obtains stable recognizability within structural constraint.

Chapter 11

Bird Flocks: The Dynamic Radiance of Centerless Synchrony

Core Proposition of This Chapter
The whole movement of a bird flock is not a central command, but dynamic radiance formed by local individuals within constraints of speed, distance, direction, and collision avoidance.

11.1 Why a Bird Flock Looks Like a Whole

When birds fly in a flock, they often present a powerful sense of wholeness. They turn, divide, gather, and transform synchronously, like a flowing living shape.

But this sense of wholeness does not require a central commander. Each bird mainly responds to neighboring individuals, differences of speed, changes of direction, and environmental pressure. Wholeness comes from relations, not from a single subject.

11.2 Distance, Speed, and Collision Avoidance

A bird flock can be maintained because individuals can neither draw infinitely close nor scatter completely. Too close, and they collide; too far, and they disperse; too slow in response, and they break; too excessive in response, and they become chaotic.

Distance, speed, direction, and collision avoidance constitute the basic topology of the bird flock. This topology is not a static diagram, but a continuously renewed relational field.

11.3 Dynamic Radiance

The special feature of a bird flock's radiance is that it is almost entirely dynamic. Ant trails can leave scent traces; termite mounds can condense into material structures; but the bird flock formation exists in movement.

This shows that a vessel need not always be static. As long as a relation can bear form, boundary, and function, it can become a vessel. The bird flock formation is a vessel of movement.

11.4 Division and Reorganization

A bird flock may divide under predatory pressure, changes in wind direction, or individual differences. Yet division does not necessarily terminate collective radiance; new local formations may rapidly reorganize.

The bird flock therefore displays a highly elastic collective vessel. Its boundary is variable, its form is fluid, and its stability comes from continual revision rather than fixed immutability.

11.5 The Implication of Centerless Order

The bird flock shows most clearly that centerlessness is not disorder. Order can arise from coordination among local responses rather than from command from above.

This has implications for understanding complex systems, network organizations, and even civilizational institutions: many higher-order forms of order are not produced by a center, but manifest within relational constraint.

Chapter Summary

This chapter uses the bird flock to explain dynamic radiance. The whole turning of a flock is not a central command, but a movement structure formed by individuals through distance, speed, visual range, collision avoidance, and neighbor response.

The bird flock as vessel is not a fixed object, but a relational form in motion. It continually manifests through synchrony, division, re-aggregation, and directional adjustment, displaying the dynamic stability of centerless order.

Chapter 12

Fish Schools: Defense, Direction, and Fluid Boundaries

Core Proposition of This Chapter
The boundary of a fish school is not a fixed wall, but a fluid boundary continually formed through currents, danger, neighbor response, and directional synchrony.

12.1 The Specificity of Group Order in Water

Fish schools live in a fluid environment, and therefore their boundaries more clearly appear as fluid boundaries. Water currents, light, predators, individual speed, and neighbor distance jointly shape the form of the school.

A fish school is not simple aggregation. It forms a defensive collective vessel through directional synchrony, distance maintenance, and rapid turning.

12.2 Predatory Pressure and Group Boundary

Predatory pressure often makes the boundary of a fish school clearer. Danger is not merely an external disturbance; it changes the space of admissibility, making certain dense formations, evasive directions, and turning patterns easier to bear.

This shows that threat can also participate in manifestation. Threat does not create the group, but by constraining paths, it opens certain group forms.

12.3 The Defensive Collective Vessel

The defense of a fish school is not a simple addition of individual defenses. The whole boundary, synchronized turning, and changes in density jointly produce a defensive effect.

The individual remains vulnerable, but the collective vessel changes the structure faced by the predator. The group does not make the individual disappear; it reorganizes the individual's position of risk.

12.4 Fluidity and Reorganization

Fish schools often break apart in an instant and reorganize in an instant. This fluidity does not show that they have no structure; it shows that their structure is highly dynamic.

Fish schools remind us that stability is not rigidity. Sometimes the strongest group stability arises precisely from rapidly variable boundaries.

12.5 Boundary as Process

The boundary of a fish school is not a line drawn once and for all, but a continuously generated process. It changes with speed, direction, danger, and current.

Thus, boundary is not a static limit, but part of the process of manifestation itself. Without continuously generated boundaries, the fish school could not appear as a defensive whole.

Chapter Summary

This chapter uses fish schools to explain fluid boundaries. The boundary of a fish school is not a fixed wall-like contour, but a defensive structure continually formed in water currents, predatory pressure, neighbor synchrony, and directional adjustment.

Fish schools show that the collective vessel can be highly temporary and yet effective. It contracts under danger, extends along paths, breaks under disturbance, and rapidly reorganizes into a new fluid order.

Chapter 13

Herds and Packs: Memory, Hierarchy, Migration, and Protective Structures

Core Proposition of This Chapter
Herds and packs show that the collective vessel can be maintained not only by instinct and local feedback, but also through memory, roles, learning, and intergenerational transmission.

13.1 How Herds and Packs Differ from Insect Societies

Many herds and packs differ from insect societies. They often possess stronger individual learning capacities, more complex kinship relations, more obvious role changes, and richer social memory.

This makes them a transitional field between natural groups and civilizational groups. They are not yet civilizations, but they already display memory, protection, hierarchy, migration routes, and intergenerational transmission.

13.2 Migration Routes as Collective Memory

A migration route is not merely a spatial path, but a vessel of collective memory. The route is maintained through repeated action, environmental markers, the experience of older individuals, and group following.

Individuals may die, but the route can continue to manifest within the group. This already approaches a prehistoric form of civilizational memory: memory is stored not only in individual brains, but also in structures of collective action.

13.3 Hierarchy, Roles, and Protection

Many herds and packs possess hierarchy, territory, protection of young, cooperative hunting, or divisions of vigilance. These structures are not simple rankings of strength, but the role topology of the collective vessel.

Role topology allows the group to organize risk and resources. Strong individuals, older individuals, young individuals, and peripheral individuals occupy different functional positions, and these positions jointly determine how the group acts.

13.4 Learning and Intergenerational Transmission

Learning in herds and packs allows the collective vessel no longer to depend only on immediate stimulus. Experience can be transmitted to the next generation through imitation, following, vigilance, and repeated paths.

Although this transmission is not yet the texts and institutions of human civilization, it already displays an embryonic historicity: past structures of action can influence future spaces of action.

13.5 The Threshold from Herds and Packs to the Civilizational Vessel

Herds and packs display some preconditions of civilization: memory, cooperation, roles, migration, protection, learning, and intergenerational transmission. Yet they still lack fully linguistically articulated history, institutionalized rules, abstract values, and technological accumulation.

Thus, herds and packs are not miniature civilizations, but a higher-order natural basis before the appearance of the civilizational vessel.

Chapter Summary

This chapter uses herds and packs to show how the collective vessel begins to approach social structure. Migration routes, protection of young, hierarchical relations, role differentiation, and the transmission of experience mean that such groups are no longer merely synchronized motion; they possess preliminary collective memory and organizational continuity.

Herds and packs provide a prehistoric model for the civilizational vessel. They have not yet entered linguistically articulated history and reflective institutions, yet they already show how memory, learning, transmission, and protective structure can enhance the stability of collective manifestation.

Part V

Collective Intelligence, the Boundary of Value, and the Closure of Manifestational Theory

The task of Part V is to develop an applied analysis of manifestational theory around collective intelligence, the boundary of value, and the closure of manifestational theory. The following chapters explain, through concepts, levels, cases, and boundaries of value, how group behavior obtains stable recognizability within structural constraint.

Chapter 14

Collective Intelligence Is Not a Collective Soul

Core Proposition of This Chapter
Collective intelligence is not a soul hidden behind the group, but a higher-order operable order formed by multiple individuals within structural constraint.

14.1 Why the Phrase "Collective Intelligence" Is Easily Misleading

The phrase "collective intelligence" is attractive because it grasps the fact that whole order exceeds individual capacity. But it is also easily misleading, because people may understand intelligence as the mental activity of some hidden subject.

This book preserves the higher-order character of group structure, but does not mystify it. Collective intelligence is not a collective soul, but the functional performance of the collective vessel.

14.2 Higher-Order Order Is Not a Higher-Order Subject

A pattern can stand above the individual without becoming a subject. An ant trail is higher than a single ant; a bee colony is higher than a single bee; a bird flock formation is higher than a single bird. But this does not mean that a unified consciousness stands behind them.

Higher-order explanatory power and subjectivity are two different things. Manifestational theory acknowledges the former and refuses to hastily infer the latter.

14.3 Collective Will Is Only the Functional Name for the Persistence of Difference

If we say that a group has "will," this can only be said in a very strict sense: certain differences persist within the group structure. Path persistence, directional persistence, division-of-labor persistence, and migration-route persistence are not psychological will, but structural functions.

This definition avoids two errors: one is reducing group behavior to fragments of individual action; the other is mystifying group behavior into a whole soul.

14.4 A Demystified Definition of Collective Intelligence

This book suggests defining collective intelligence as follows: a higher-order operable order formed by multiple living individuals within a shared environment, local signals, feedback mechanisms, and a space of admissibility.

This definition acknowledges the effectiveness of group behavior while limiting over-interpretation. The collective vessel can do things that appear intelligent, but it need not possess an "I."

14.5 Restrained Holism

The position of this book can be called restrained holism. It acknowledges that whole structures are real and effective, but does not reify the whole; it acknowledges that group patterns cannot be ignored, but does not deny individual mechanisms.

This restraint is crucial for manifestational theory. Without restraint, the theory would slide into mysticism; without wholeness, the theory would regress into fragmented reduction.

Chapter Summary

This chapter demystifies collective intelligence. To acknowledge that a group possesses higher-order order does not mean that a hidden soul or supersubject exists behind the group. More precisely, collective intelligence is operable order formed by local rules within structural constraint.

The book therefore preserves the explanatory power of the whole while rejecting the myth of the whole. Collective will, collective radiance, and the collective vessel are all functional concepts used to describe difference, form, and bearing relations, not new mysterious entities.

Chapter 15

Natural Stability Is Not the Good

Core Proposition of This Chapter

Animal and insect groups can display natural forms of sustainable manifestation, but they cannot directly yield the good in the ethical sense.

15.1 Why Group Behavior Raises the Question of Value

When we see bee colonies maintaining themselves, bird flocks coordinating, fish schools defending themselves, and herds protecting their young, it is easy to treat natural stability as good. Yet locust swarms are also stable; predation is also effective; parasitism can also continue.

The stability of nature cannot directly become an ethical judgment. The value theory of manifestational ontology must avoid equating "able to exist," "able to expand," or "able to adapt" directly with "good."

15.2 The Difference among Survival, Reproduction, Expansion, and the Good

Survival is the maintenance of oneself, reproduction is the continuation of a type, and expansion is the enlargement of range. All can be important modes of natural manifestation, but they do not automatically become ethical goods.

A structure can live for a long time while maintaining itself by destroying other bearing conditions; a group can expand while consuming future possibilities. Therefore, the good cannot be defined as natural success.

15.3 What Natural Groups Teach Value Theory

Natural groups are not the judges of value theory, but they are important material for value theory. They allow us to see the real effects of structure, boundary, feedback, and bearing conditions, and they also allow us to see the danger of destructive manifestation.

Between natural groups and human value, reflection must intervene. Human beings should not simply imitate nature; rather, they should understand the structural conditions of natural manifestation and judge at a higher level which structures deserve continuation.

15.4 Against the Naturalistic Fallacy

This book explicitly opposes deriving ethical norms directly from natural facts. How animal groups act cannot automatically tell human beings how they ought to act.

Manifestational theory provides structural understanding, not the worship of nature. Natural stability can help us understand bearing, but it cannot decide the good for us.

15.5 The Good as the Structural Enhancement of Complex Manifestation

Within the theoretical framework of this book, the good is closer to a structural direction that enhances non-destructive, sustainable, complex manifestation. This definition is stricter than natural success.

Even if a group structure is stable, if it mainly maintains itself by devouring the future, destroying bearing conditions, and compressing other forms of complex manifestation, it cannot be directly called good.

Chapter Summary

This chapter draws the boundary between natural order and value judgment. Natural selection, adaptive success, reproductive expansion, and group stability cannot be directly equated with the good in the ethical sense.

Animal and insect group behavior can display how manifestation becomes stable, but it cannot complete value judgment for us. The question of the good must return to whether a structure enhances the bearing capacity of non-destructive, sustainable, complex manifestation.

Chapter 16

From the Collective Vessel to the Civilizational Vessel

Core Proposition of This Chapter
Human civilization is not a miracle detached from natural group order, but the historical unfolding of the collective vessel at the levels of language, memory, institution, technology, and value.

16.1 Continuity and Break

There is both continuity and break between animal groups and human civilization. The continuity lies in the fact that both require boundaries, memory, roles, signals, cooperation, and bearing structures. The break lies in the fact that civilization introduces linguistically articulated history, institutionalized rules, technological accumulation, reflective value, and transmissible texts.

Civilization therefore does not appear out of nothing, nor is it a simple enlargement of animal groups. It is the historicization, symbolization, and value-formation of the collective vessel at a higher level.

16.2 How Collective Memory Becomes History

The migration route of a herd is collective memory, but it is not yet history. History requires memory to be narrated, interpreted, contested, preserved, and transmitted. The civilizational vessel transforms collective memory into a reflective temporal structure.

This is why civilizations have myths, rituals, monuments, texts, and institutions. They are all vessels that allow collective radiance to cross the lives of individual persons.

16.3 How a Nest Becomes a City

A nest is a materialized collective vessel, whereas a city is a civilized collective vessel. A city bears not only habitation, but also law, exchange, identity, technology, memory, and power relations.

The movement from a termite mound to a human city is not a linear evolution, but a leap in levels of bearing: material structure enters symbolic structure; spatial boundary enters institutional boundary; group maintenance enters historical continuation.

16.4 From Natural Groups to Civilizational Value

Natural groups show us how order manifests under centerless conditions, while civilization forces us to face a more difficult question: which orders deserve preservation? Which radiances ought to continue? Which vessels, although powerful, consume the future?

Thus, this book finally returns to value: the civilizational vessel cannot pursue stability alone; it must also ask about the direction of stability.

16.5 The Relation between This Short Monograph and Manifestational Ontology

This book does not start a new project outside the main theory; it brings manifestational ontology into natural living systems. It shows that manifestational theory applies not only to civilization, AI, and value, but also to centerless order among animals and insects.

From the life-vessel to the collective vessel and then to the civilizational vessel, a clear theoretical line is formed: being takes shape within constraint, life maintains itself within boundary, the group manifests within relation, and civilization bears radiance within history.

Chapter Summary

This chapter connects animal groups with human civilization while preserving their difference of level. The collective vessel provides a prehistoric model for the civilizational vessel, but civilization undergoes a new leap in language, memory, institution, technology, works, and reflection on value.

From ant trails to cities, from signals to language, from nests to institutions, manifestational structures continually obtain more complex modes of bearing. Civilization is not a miracle separated from natural order, but the reorganization and elevation of collective manifestation on the historical scale.

Conclusion

The Group Is a Life-Order Manifesting under Constraint

Beginning from the group behavior of animals and insects, this book has attempted to state a more general manifestational proposition: many forms of order are not directly produced by a central subject, nor are they secretly controlled by a mysterious whole will. They gradually manifest within local differences, structural constraint, feedback mechanisms, and a space of admissibility.

Ant trails show that paths can form temporary vessels through the persistence of difference; bee colonies show that division of labor, nests, and resource flows can form higher-order self-maintenance; termite mounds show that group behavior can materialize into architectural vessels; locust swarms show that collective manifestation can be negative and that stability is not the good; bird flocks show that dynamic relations can become vessels of movement; fish schools show that fluid boundaries can bear defense; herds and packs show that memory, roles, and migration already approach the prehistory of the civilizational vessel.

Together these cases show that group behavior cannot be fully compressed into the causal sum of individual actions. Individual mechanisms are of course important, but whole stability, recognizability, boundary, exhaustion, and reorganization require explanation at the structural level. Manifestational theory does not deny biological mechanisms; it provides a philosophical language for how such mechanisms compose higher-order forms.

At the same time, this book refuses to mystify the group. Collective intelligence is not a collective soul, collective will is not a psychological subject, and collective radiance is not mystical energy. They are functional concepts used to describe how difference persists, how form becomes recognizable, how vessels bear, and how structures become exhausted and reorganized.

Animal and insect group behavior therefore becomes an important natural field of application for manifestational ontology. It shows that many forms in the world do not merely occur, but take shape within constraint; they are not completed once and for all, but maintained through time; they are not eternally stable, but continue to appear through exhaustion and reorganization.

A group is not commanded into being. A group takes shape within constraint. When multiple life-vessels together open a space of admissibility, collective radiance begins to appear.

Appendix A

Glossary of Core Terms

Term	Definition
Collective radiance	The stable and recognizable whole pattern formed by multiple living individuals within structural constraint.
Collective vessel	A higher-order bearing structure formed by multiple life-vessels within a shared environment, local signals, feedback mechanisms, and boundary conditions.
Life-vessel	A manifestational structure capable of maintaining its own boundary, exchanging energy and information, resisting collapse, and participating in its own continuation.
Persistence of difference	A local difference does not immediately disappear, but is continuously reinforced through signals, environment, or feedback.
Collective admissibility	The range of group forms that a living system can stably bear under given conditions of body, environment, signal, and feedback.
Dynamic vessel	A vessel whose primary mode of bearing is a relation of movement rather than a fixed object, such as a bird flock formation or a fish school boundary.
Materialized collective vessel	A bearing form in which group behavior condenses into material structure, such as a termite mound or beehive.
Negative collective manifestation	Collective manifestation that has a stable group structure but primarily operates through consuming, devouring, or destroying bearing conditions.
Collective exhaustion	The loss of the original bearing capacity of a collective vessel due to the depletion of resources, signals, boundaries, health, or environmental conditions.
Collective reorganization	The formation of a new structure by local differences under new conditions after an old collective vessel has collapsed or become exhausted.

Appendix B

Six-Level Analytic Table of Collective Manifestation

Level	Correspondence in Group Behavior	Example
L0 Unmanifest precondition	Resources, threats, terrain, temperature, individual density, bodily capacities, and perceptual background	The ant-colony environment before food appears; the state of a fish school before a predator approaches
L1 Will / persistence of difference	A local path, direction, signal, or behavioral deviation is continuously reinforced	A pheromone path is strengthened; a change of direction propagates through a bird flock
L2 Radiance	A group pattern acquires stable recognizability	Ant trail, fish school boundary, bird flock formation, bee-colony division of labor
L3 Topology and admissibility	Individual relations, distance, signal paths, role relations, and spatial boundaries select stable forms	Collision-avoidance distance in bird flocks; role distribution in bee colonies; air-flow structure in termite mounds
L4 Vessel	A group structure acquires a bearing function	Beehive, termite mound, ant-trail network, migration route
L5 Collapse and reorganization	A path disappears, a formation breaks, a group relocates its nest or swarms, or the group reorganizes	A new path appears after an ant trail is exhausted; a fish school reorganizes after being startled

Appendix C

Case Reference Table

Case	Main Explanatory Focus
Ant colony	Persistence of difference; path becoming vessel; environmental memory
Bee colony	Division of labor; nest boundary; higher-order self-maintenance
Termites	Materialized collective vessel; architectural manifestation
Locust swarm	Negative manifestation; stability is not the good
Bird flock	Centerless synchrony; dynamic radiance
Fish school	Fluid boundary; defensive collective vessel
Herds and packs	Migration memory; role topology; intergenerational transmission

Appendix D

Typical Misreadings and Responses

Misreading 1: This book claims that animal groups have souls.

Response: No. The collective vessel is structural manifestation, not a soul-entity.

Misreading 2: This book denies the mechanisms studied by ethology.

Response: No. Causal mechanisms explain concrete occurrence; manifestational theory explains how whole patterns stably take shape.

Misreading 3: This book equates the strong in nature with the good.

Response: No. Natural stability, adaptive success, and ethical goodness must be distinguished.

Misreading 4: This book mystifies insect societies.

Response: No. The book is precisely an attempt to demystify collective intelligence.

Misreading 5: This book reduces human civilization to animal groups.

Response: No. Animal groups provide a prehistoric model, while civilization undergoes a leap in language, memory, institution, technology, and value.

References and Further Reading

The following works provide the main empirical and theoretical background for the book's discussion of group behavior, self-organization, social insects, collective motion, and complex systems. This book is a philosophical text of structural explanation; the bibliography is provided as empirical and theoretical background.

1. Bonabeau, E., Dorigo, M., & Theraulaz, G. (1999). Swarm Intelligence: From Natural to Artificial Systems. Oxford University Press.

2. Camazine, S., Deneubourg, J.-L., Franks, N. R., Sneyd, J., Theraulaz, G., & Bonabeau, E. (2001). Self-Organization in Biological Systems. Princeton University Press.

3. Couzin, I. D., & Krause, J. (2003). Self-organization and collective behavior in vertebrates. Advances in the Study of Behavior, 32, 1-75.

4. Couzin, I. D., Krause, J., Franks, N. R., & Levin, S. A. (2005). Effective leadership and decision-making in animal groups on the move. Nature, 433, 513-516.

5. Gordon, D. M. (2010). Ant Encounters: Interaction Networks and Colony Behavior. Princeton University Press.

6. Hölldobler, B., & Wilson, E. O. (1990). The Ants. Harvard University Press.

7. Hölldobler, B., & Wilson, E. O. (2009). The Superorganism: The Beauty, Elegance, and Strangeness of Insect Societies. W. W. Norton.

8. Reynolds, C. W. (1987). Flocks, herds, and schools: A distributed behavioral model. Computer Graphics, 21(4), 25-34.

9. Seeley, T. D. (2010). Honeybee Democracy. Princeton University Press.

10. Sumpter, D. J. T. (2010). Collective Animal Behavior. Princeton University Press.

11. Turner, J. S. (2000). The Extended Organism: The Physiology of Animal-Built Structures. Harvard University Press.

12. Vicsek, T., Czirók, A., Ben-Jacob, E., Cohen, I., & Shochet, O. (1995). Novel type of phase transition in a system of self-driven particles. Physical Review Letters, 75(6), 1226-1229.

13. Wilson, E. O. (1971). The Insect Societies. Harvard University Press.

14. Wilson, E. O. (1975). Sociobiology: The New Synthesis. Harvard University Press.